

The cellular mechanism of the higher thermal sensitivity of methanogenesis relative to respiration

Gabriel Yvon-Durocher

Living Systems Institute, University of Exeter, Stocker Road, Exeter EX4 4QD, UK

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Abstract

The thermal sensitivity of methane production is higher than that of respiration, and remarkably this holds consistently from individual methanogen cultures to whole ecosystems¹ — implying the pattern originates at the cellular scale. Yet the cellular mechanism is unknown, and more generally it is unresolved whether the activation energy (E_a) of a metabolic rate reflects one rate-limiting enzyme, an aggregate of the proteome, or a biophysical constant. We address this with mechanistic enzyme- and temperature-constrained genome-scale models (etc-GEMs) — the first for their taxa — of a respiring bacterium (*Escherichia coli*) and a hydrogenotrophic methanogen (*Methanococcus maripaludis*), each calibrated to a measured growth thermal performance curve. Quantifying E_a with the window-independent Sharpe–Schoolfield E , both models reproduce the observed value (*E. coli* 0.68 eV vs observed 0.56, on the “universal” ~0.65 eV benchmark; *M. maripaludis* 1.06 vs 1.04), and the two calibration posteriors are cleanly separated — reproducing the empirical pattern at the cellular scale. A non-circular, control-weighted decomposition shows the +0.38 eV difference is **structural, not thermodynamic**: the methanogen’s enzymes are on average *less* temperature-sensitive (0.51 vs 0.87 eV), but its growth control concentrates on the high- E_a methanogenesis backbone (+0.34 vs +0.11 eV), and its constant-ribosome allocation strategy lacks the proteome buffer that lowers *E. coli*’s E_a (+0.00 vs –0.13 eV). Both terms are grounded in each organism’s own measured allocation physiology. The higher thermal sensitivity of methanogenesis is thus a property of metabolic architecture and resource allocation — the same control-weighted aggregation acting on a narrow, unbuffered energy backbone — providing a concrete cellular origin for a pattern that propagates across scales.

1 Introduction

The temperature dependence of biological rates is one of the most general regularities in biology. Across taxa and levels of organisation, metabolic rates increase with temperature in an approximately Boltzmann–Arrhenius fashion, and the fitted **activation energy** E_a — the slope of $\ln(\text{rate})$ against inverse temperature over the rising limb — clusters near a “universal” value of ~0.6–0.7 eV^{2,3}. This apparent constancy underpins metabolic scaling theory and is widely used to predict how biological processes respond to warming. It also raises an unresolved mechanistic question: is an organism’s E_a set by a single rate-limiting enzyme whose temperature response the organism inherits, by an aggregate over many enzymes, or by a shared thermodynamic constraint? The distinction matters, because only a mechanistic account can explain the systematic *departures* from the universal value that are also observed.

The most consequential such departure concerns methane.¹ showed empirically that the thermal sensitivity of methane flux is substantially higher than that of respiration (and of photosynthesis), and — critically — that this ordering is **consistent from individual methanogen cultures through microbial communities to whole ecosystems**. Cross-scale consistency of an activation energy is a strong clue: it implies that the pattern is not an emergent property of ecological aggregation but originates at the **cellular and enzymatic scale** and propagates upward. What that cellular mechanism is, however, has remained unknown. Why should the growth of a methanogen be more thermally sensitive than that of a respiring bacterium — is it a more temperature-sensitive enzymatic machinery, or something about how that machinery is organised and deployed?

Answering this requires a model in which the organism-level E_a can be traced back to individual enzymes. Enzyme- and temperature-constrained genome-scale models (etc-GEMs) provide exactly this: they impose a temperature-dependent turnover on every enzyme and a shared proteome budget, and predict the whole growth thermal performance curve (TPC) from enzyme-constrained flux balance analysis. Building on a yeast thermal-etcGEM framework⁴, we construct etc-GEMs — to our knowledge the first for either taxon — of *E. coli* (aerobic respiration, a complete electron transport chain) and *M. maripaludis* (hydrogenotrophic methanogenesis, whose *only* energy-conserving pathway is the Wolfe-cycle backbone terminating in the notoriously slow methyl-coenzyme-M reductase, Mcr). We calibrate each to a measured growth TPC, quantify E_a with a window-independent metric, and decompose it non-circularly into a control-weighted aggregate of the per-enzyme activation energies plus named physiological couplings. The answer, previewed here, is that the higher thermal sensitivity of methanogenesis is **structural**: not a more temperature-sensitive enzyme set, but the concentration of growth control on a single high-sensitivity energy backbone together with the absence of the proteome-allocation buffer that dampens the respiring bacterium’s E_a .

2 Results

2.1 Two etc-GEMs reproduce two separated activation energies

Each etc-GEM was calibrated by Bayesian inference to its organism’s measured growth TPC — *E. coli* on rich Brain–Heart–Infusion broth over 7–46 °C⁵, and *M. maripaludis* on H₂/CO₂ defined medium over 15–52 °C⁶. Both reproduce their data: the posterior-predictive median and 90% band pass through the observed points (Fig. 1, left and centre), and the two curves differ visibly in the steepness of their rising limbs. Propagating each calibration posterior through the Sharpe–Schoolfield fitter (Methods) yields the posterior distribution of E_a for each organism. These are **cleanly separated** — *M. maripaludis* at ~1.0 eV and *E. coli* at ~0.6 eV, with non-overlapping 90% credible intervals ([0.88, 1.16] versus [0.60, 0.80]; Fig. 1, right). The empirical pattern of¹ — higher thermal sensitivity for methanogenesis than for respiration — is thus reproduced at the cellular scale by two independent mechanistic models, and is robust to Bayesian parameter uncertainty rather than an artefact of a point estimate. The remainder of the paper asks *why*.

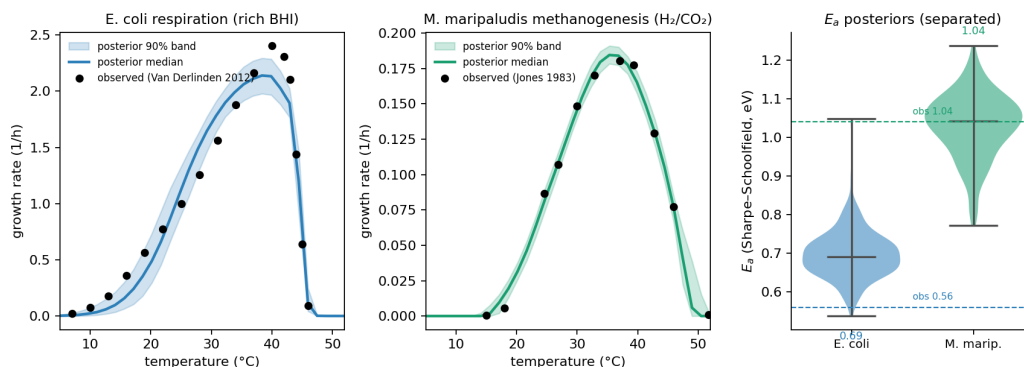


Fig. 1 | Two organisms, two activation energies, reproduced mechanistically. Calibrated (Bayesian) growth thermal performance curves for *E. coli* respiration (rich BHI) and *M. maripaludis* methanogenesis (H₂/CO₂): posterior-predictive median and 90% band with the observed data^{5,6}. Right: the Sharpe–Schoolfield E_a posteriors, propagated from the calibration draws, are separated (methanogen ~1.0 vs *E. coli* ~0.6 eV; dashed lines, observed values).

2.2 Both models predict the observed activation energy

Before decomposing the difference we establish that the modelled E_a is a genuine prediction. This test requires a stable metric, because the naive Boltzmann–Arrhenius slope is not one: the rising limb is curved (enzyme kinetics are not log-linear in inverse temperature), so the fitted slope depends on the arbitrary temperature window. Recomputed under five window conventions, the slope E_a of these very curves spans **0.42–1.41**

eV per curve — larger than the cross-organism differences we seek to explain, and enough that under one window the two organisms appear comparable while under another they are far apart (Table 1). We therefore quantify E_a throughout as the window-independent **Sharpe–Schoolfield E^7** , fit to the entire curve shape (Methods); the windowed slope is retained only as a deprecated cross-check.

Under this metric the models agree with the data. The *E. coli* model gives $E_a = 0.68$ eV against an observed 0.56 eV, sitting on the “universal” ~ 0.65 eV metabolic benchmark; the *M. maripaludis* model gives 1.06 eV against an observed 1.04 (Table 2). Because the per-enzyme activation energies are model *inputs* (below), this agreement is a prediction of the organism-level value, not a fitted output — and the methanogen genuinely sits ~ 1.6 -fold above the respiring bacterium.

Table 1: The naive Arrhenius-slope E_a is window-dependent, the Sharpe-Schoolfield E is not. Slope E_a (eV) for the model and observed curves under five window conventions (range shown), versus the window-independent Sharpe-Schoolfield E . The per-curve slope spread (0.42–1.41 eV) exceeds the cross-organism differences; the Sharpe-Schoolfield E is stable and makes model and observed agree.

curve	slope (10–95%)	slope range	Sharpe–Schoolfield E
<i>E. coli</i> model	0.96	0.76–1.56	0.68
<i>E. coli</i> observed	0.64	0.55–0.96	0.56
methanogen model	1.17	0.89–1.78	1.02
methanogen observed	0.66	0.66–2.07	1.04

2.3 The difference is structural: control weighting and allocation, not enzyme thermodynamics

To find what sets each E_a we decompose it, by metabolic control analysis, into a control-weighted aggregate of the per-enzyme activation energies and four named physiological couplings (Methods, Equation 2). The framing is deliberately non-circular: the per-enzyme activation energies E_i are grounded model inputs, so we do not ask “what is E_a ” but how the organism-level value aggregates them and departs from their naive mean. Fig. 2 and Table 2 give the result for both organisms on the same terms.

The central finding is counter-intuitive. The methanogen’s enzymes are, on average, *less* temperature-sensitive than the respiring bacterium’s: the unweighted mean of the per-enzyme activation energies is 0.51 eV in *M. maripaludis* versus 0.87 eV in *E. coli*. Intrinsic enzyme thermodynamics alone would therefore predict the *opposite* ordering to the one observed. The higher methanogen E_a arises instead from how growth control is distributed and how the proteome is allocated.

First, growth control concentrates on the high- E_a methanogenesis backbone. In a hydrogenotroph, energy metabolism *is* growth: essentially all catabolism flows through the Wolfe cycle, so a small set of energy enzymes — formylmethanofuran dehydrogenase (Fwd), the methyltransferase (Mtr), Mcr, the methylene-tetrahydromethanopterin reductase (Mer), the heterodisulfide reductase (Hdr) and the hydrogenases — carries **89% of the control term**, and these enzymes are themselves high- E_a (mean 0.77 eV against a proteome mean of 0.51). This concentration lifts the effective enzyme activation energy by **+0.34 eV**. In *E. coli*, by contrast, growth control is spread across a broad biosynthetic proteome, and the analogous control-weighting term adds only **+0.11 eV**.

Second, the methanogen lacks the proteome-allocation buffer that lowers *E. coli*’s E_a . In *E. coli*, the Scott-type growth law reallocates proteome toward ribosomes as growth accelerates, which shrinks the metabolic sector at higher temperature and pulls the organism E_a down by **−0.13 eV**. *M. maripaludis*, which maintains a **constant ribosomal proteome fraction** and full methanogenesis capacity across growth rate⁸, applies no such buffer: its allocation term, measured on its own sectorized model, is **+0.00 eV**. Both allocation terms are therefore grounded in each organism’s own measured allocation strategy, making the comparison symmetric rather than an artefact of model structure.

Together these two structural terms — the control-weighting difference (+0.24 eV) and the allocation asymmetry (+0.13 eV) — account for essentially the entire +0.38 eV gap in organism E_a . The two remaining

terms nearly cancel: the methanogen’s lower average enzyme activation energy (-0.36 eV) is offset by a larger **aggregation** term ($+0.39$ eV), the tendency of its narrow, synchronised control on a few backbone enzymes to steepen the composite growth curve where *E. coli*’s broad control flattens it. We note that although the aggregation term is the largest single contribution, it is also the most sensitive to the details of the metric; the robust conclusion — that the difference is set by control weighting and allocation — does not depend on it.

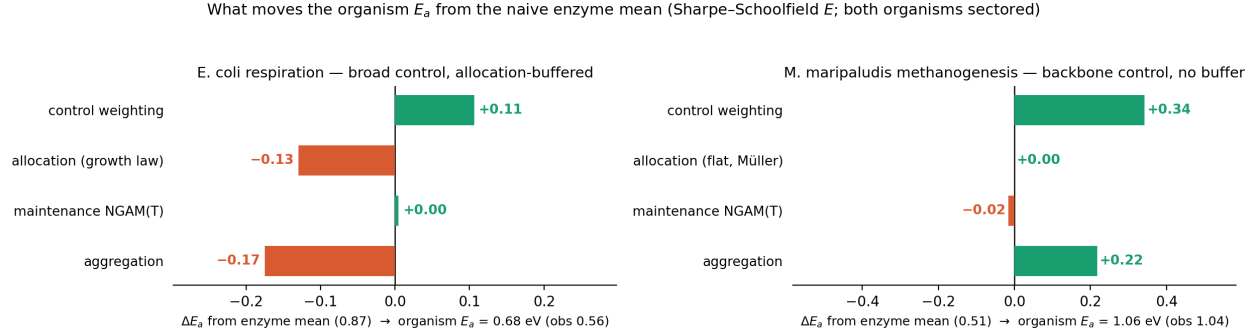


Fig. 2 | What moves the organism E_a away from the naive enzyme mean. Signed contribution of each named term (Sharpe-Schoolfield E ; both organisms carrying a proteome-sector layer), sign-coloured. The higher methanogen E_a is driven by control weighting ($+0.11$ vs $+0.34$ eV) and the allocation asymmetry (-0.13 vs $+0.00$ eV), not by higher intrinsic enzyme temperature-sensitivity (naive mean 0.87 vs 0.51 eV).

Table 2: Symmetric cross-organism decomposition of the Sharpe-Schoolfield activation energy. Both organisms carry a proteome-sector/growth-law layer, so the allocation term is measured for each (*E. coli* Scott growth law; *M. maripaludis* constant-ribosome flat law). The $+0.38$ eV difference is set by control-weighting and allocation, not by a higher intrinsic enzyme E_a (naive mean 0.87 vs 0.51). Both models match the observed SS- E .

term	<i>E. coli</i> (rich BHI)	<i>M. maripaludis</i> (H2/CO2)
organism SS- E E_a (eV)	0.6784	1.0558
observed SS- E (eV)	0.558	1.042
model – observed	0.12	0.014
Topt (°C)	38.4	35.3
CTmax (°C)	46.0	48.8
naive (unweighted) enzyme- E mean	0.8716	0.5118
+ control weighting	0.1063	0.3427
+ allocation (measured; <i>E. coli</i> Scott / methanogen flat)	-0.1295	0
+ maintenance NGAM(T)	0.0049	-0.0164
+ aggregation (heterogeneity/flattening)	-0.1749	0.2177
backbone fraction of control term	n/a (broad)	0.887
sum C_i	0.556	1.189

2.4 The ordering is robust to the uncertain Mcr turnover

The turnover number of the terminal methanogenesis enzyme, Mcr, is uncertain for a mesophile — the literature spans roughly $3\text{--}294\text{ s}^{-1}$. Because Mcr is a candidate rate-controller, we swept its k_{cat} across this range and repeated the analysis (Table 3). The organism E_a remains between 1.01 and 1.27 eV throughout, always well above the *E. coli* value of 0.68, and the methanogenesis backbone carries 83–103% of the control term across the whole range. The ordering and the backbone-control finding are therefore robust to this uncertainty. What does change is the identity of the single leading enzyme: Mcr dominates when it is slow (a low turnover makes it expensive in enzyme mass and thus high in control), Fwd leads when Mcr is fast, and

the two co-lead at the calibrated value. The classic expectation that Mcr specifically sets the methanogen E_a therefore holds only when Mcr is slow; the robust statement is that the *energy backbone as a whole*, not any single enzyme, carries the high thermal sensitivity.

Table 3: Sensitivity to the Mcr turnover number (3-294/s). The organism SS-E stays well above *E. coli*’s 0.68 and the backbone fraction of the control term stays near-total across the range; only the leading enzyme shifts (Mcr when slow, Fwd when fast).

Mcr kcat (1/s)	organism SS-E (eV)	backbone fraction of control	leading enzymes
3	1.2743	1.026	Mcr(1.605);Fwd(0.092);CODH(-0.065)
10	1.0105	0.986	Mcr(0.876);Fwd(0.167);CODH(-0.039)
30	1.0651	0.925	Mcr(0.488);Fwd(0.277);Mtr(0.059)
58.9	1.0558	0.887	Fwd(0.337);Mcr(0.304);Mtr(0.070)
100	1.0659	0.857	Fwd(0.395);Mcr(0.211);Mtr(0.079)
294	1.0171	0.833	Fwd(0.427);Mtr(0.086);Mer(0.079)

3 Discussion

The higher thermal sensitivity of methanogenesis relative to respiration¹ has a concrete cellular mechanism, and it is not the intuitive one. The methanogen’s enzymes are not individually more temperature-sensitive — on average they are less so. Instead the organism-level activation energy is a **control-weighted, allocation-modulated aggregate** of the per-enzyme activation energies, and two structural features of methanogenic physiology raise it. Because a hydrogenotroph conserves energy through a single narrow pathway, growth control concentrates almost entirely on that high-activation-energy energy backbone, rather than being diluted across a broad biosynthetic proteome as in a respiring heterotroph. And because *M. maripaludis* maintains a constant ribosomal fraction and full methanogenesis capacity across growth rate⁸, it lacks the growth-law proteome reallocation that buffers *E. coli*’s activation energy downward. Each term is grounded in the organism’s own measured allocation strategy, so the comparison reflects biology rather than a modelling choice.

This reframes the “universal” metabolic activation energy. Its approximate constancy need not imply a conserved enzyme or a fixed biophysical constant; the same control-weighted-aggregation mechanism, acting on a **different controlling set** and modulated by a **different allocation strategy**, produces systematically different organism-level values. A respiring bacterium lands near the benchmark because its control is broad and allocation-buffered; a methanogen sits well above it because its control is narrow and unbuffered. The corollary is a link, to our knowledge not previously drawn, between **proteome-allocation strategy and thermal sensitivity**: how an organism partitions and couples its proteome is a determinant of how steeply its growth responds to temperature. This connection is significant precisely because of the cross-scale consistency that motivated the question — if the cellular mechanism is control concentration on a high- E_a , unbuffered energy pathway, then that mechanism is what propagates the methanogenesis signal from cultures up to the ecosystem fluxes of¹.

Several caveats bound these conclusions. The aggregation term, though the largest single contribution to the cross-organism difference, is the most sensitive to the definition of the activation-energy metric; we have therefore rested the mechanistic conclusion on the control-weighting and allocation terms, which alone account for the observed gap. The mesophilic turnover of Mcr is the dominant single-parameter uncertainty: the ordering and the backbone-control result are robust across its plausible range, but the identity of the single most rate-controlling enzyme is not, and a direct measurement of mesophilic Mcr k_{cat} would sharpen the attribution. The per-enzyme activation energies are grounded inputs (from macromolecular rate theory, sequence-based optima, a melting proteome, and — for the methanogen — a deep-learning turnover predictor corrected against literature kinetics), so the result is an attribution given those inputs rather than an ab-initio

prediction of the numerical value. Finally, this is a two-organism comparison; the mechanism is a hypothesis that a broader survey should test.

The natural next test is the third process in the¹ ordering. Photosynthesis has an even lower thermal sensitivity than respiration, and a cyanobacterial etc-GEM — with its light-driven energy metabolism and distinct proteome economy — would provide a demanding, directional test of whether the control-and-allocation framework generalises across all three metabolic strategies. If it does, a phenomenological “universal” activation energy will have been replaced by a mechanistic, organism-specific, and predictive account of thermal sensitivity.

4 Methods

4.1 The enzyme- and temperature-constrained genome-scale model framework

Each organism is represented by a genome-scale metabolic network solved by flux-balance analysis (FBA) with cobrapy⁹ and the Gurobi solver, augmented with an enzyme-mass budget and a temperature layer. Building on the yeast thermal-etcGEM of⁴, we extend the framework in four ways developed here: to two new taxa; with measured- and machine-learning-grounded per-enzyme parameters; with a proteome-sector partition carrying an **organism-specific** allocation law; and with an exact-likelihood Bayesian calibration and a window-independent activation-energy metric.

Enzyme constraint. Metabolic flux is limited by a shared proteome pool: each catalysed reaction i consumes enzyme mass $MW_i/k_{cat,i}$ per unit flux from a budget P_{metab} , following the GECKO^{10,11} and sMOMENT formalisms. The budget is set from independent data as $P_{metab} = \sigma f_{metab} P^{lit}$, with literature total protein P^{lit} , the measured metabolic proteome fraction f_{metab} , and the average in-vivo enzyme saturation σ (~ 0.45 , carried with its 0.4–0.5 literature range)^{12,13} — not the growth-calibrated pool, so the predicted growth-rate magnitude is emergent.

Temperature layer. Each enzyme’s turnover follows macromolecular rate theory (MMRT)¹⁴: $\ln k_{cat}(T)$ is quadratic in temperature through a heat capacity of activation ΔC_p , anchored at a per-enzyme optimum T_{opt} . Optima are predicted per enzyme from sequence with a TOME-style predictor after¹⁵ and refined, where the fit is good, by DLTKcat, a deep-learning predictor of temperature-dependent k_{cat} ¹⁶; ΔC_p is a single independent literature prior ($-4 \text{ kJ mol}^{-1} \text{ K}^{-1}$)¹⁴, deliberately not fitted, so the activation energy emerges rather than being imposed. The high-temperature collapse is set by protein denaturation: a two-state native/unfolded equilibrium gives a native fraction $f_{N,i}(T)$ keyed on a per-enzyme melting temperature T_m , and the effective per-flux cost is $MW_i/(k_{cat,i}(T) f_{N,i}(T))$. A temperature-dependent non-growth maintenance demand, NGAM(T), rises with temperature and rounds the optimum.

Proteome sectors and the growth law. The total protein is partitioned into three coarse sectors^{17,18} summing to one — metabolic (f_{metab}), biosynthesis/ribosomal (f_{bio}) and maintenance/housekeeping (f_{maint}) — giving the metabolic pool above and a translation cap $\propto f_{bio}$ on growth. An optional coupled growth law lets the ribosomal fraction scale with growth rate; crucially, its slope is set from **each organism’s own** proteome data rather than assumed, and this is where the two organisms differ physiologically (below).

4.2 The *E. coli* respiration etc-GEM

The base network is iML1515, the manually curated *E. coli* K-12 MG1655 reconstruction¹⁹, converted to the enzyme-constrained ecModel eciML1515 with GECKO^{10,11}. A model-quality correction was required: four reactions consume O₂ at negligible enzyme cost — two quinol/menaquinol monooxygenases and a malate:O₂ oxidoreductase and a cuprous oxidase — and form a futile O₂ cycle that short-circuits the costed respiratory chain; closing these four uncosted sinks (a documented, reversible default) routes O₂ through genuine, enzyme-costed cytochrome oxidase, at negligible cost to growth, so that control is attributed to real respiration. Per-enzyme T_{opt} and T_m were grounded from sequence/meltome predictors for $\sim 90\%$ of enzymes (the remainder set to dataset means), and the proteome-sector fractions from a growth-condition-matched measured *E. coli* proteome. The coupled growth law uses the bacterial Scott slope¹⁸, under which the ribosomal fraction rises

with growth rate. The model was validated against, and then Bayesian-calibrated to, the rich-medium (BHI) growth TPC of⁵.

4.3 The *M. maripaludis* methanogenesis etc-GEM

The base network is iMR539, the *M. maripaludis* reconstruction with explicit hydrogenotrophic methanogenesis and central electron bifurcation²⁰. It was first curated into a carbon-honest H₂/CO₂ autotroph: three biomass precursors that fed only the dispensable archaeum glycoprotein were folded out of biomass, a selenocysteinyl-tRNA recycling gap was repaired, organic-carbon leaks were closed and the pinned methane exchange freed, yielding growth with 99.98% of biomass carbon derived from CO (two vitamin cofactors lacking biosynthesis are supplied as documented trace supplements). An sMOMENT enzyme-cost layer was then parameterised with per-reaction turnovers by source priority: measured specific activities for the energy-conserving core (the electron-bifurcation hydrogenase/heterodisulfide-reductase complexes²¹, with classic biochemistry for the remaining Wolfe-cycle steps), converted to k_{cat} via the complex molecular weights; DLTkcat predictions¹⁶ for the biosynthetic bulk, corrected for a systematic archaeal underprediction with cited literature/BRENDA overrides for the highest-cost offenders and a documented turnover floor; and a dataset-mean fallback. Mcr, the terminal enzyme and the target of the mechanistic hypothesis, was sourced explicitly (mature-Mcr specific activity, giving $k_{\text{cat}} \approx 59 \text{ s}^{-1}$ with a 3–294 s^{-1} uncertainty range that is propagated in Section 2.4). The temperature layer uses the same MMRT/unfolding machinery with a documented mesophile T_m prior. Maintenance, NGAM(T), is anchored on the **measured** *M. maripaludis* maintenance energy²². The proteome-sector layer is grounded in the organism’s own allocation physiology:⁸ report that *M. maripaludis* holds a **constant** ribosomal proteome fraction across growth rate — the opposite of the bacterial growth law — so its growth-law slope is flat, giving the vanishing allocation buffer central to the comparison. The model was Bayesian-calibrated to the digitised growth TPC of⁶.

4.4 Activation energy and its decomposition

We quantify the organism-level activation energy as the **Sharpe–Schoolfield** E^7 , the rising-limb activation energy of the high-temperature-inactivation model

$$\text{rate}(T) = r_{T_{\text{ref}}} \frac{\exp[(E/k_B)(1/T_{\text{ref}} - 1/T)]}{1 + \exp[(E_h/k_B)(1/T_h - 1/T)]}, \quad (1)$$

fit to the whole calibrated TPC with a fixed reference temperature. Because E is read from the entire shape rather than an arbitrary window, it is stable where the naive Boltzmann–Arrhenius slope is not (Section 2.2), and it makes the modelled and observed values directly comparable. The same fit is applied identically to model and observed curves and to both organisms; the windowed slope is retained only as a deprecated cross-check.

We decompose the Sharpe–Schoolfield E_a , by metabolic control analysis, into a control-weighted mean of the per-enzyme rising-limb activation energies and four named couplings, relative to the naive (unweighted) enzyme mean:

$$E_a^{\text{org}} = \overline{E_i} + \underbrace{(\sum_i \hat{C}_i E_i - \overline{E_i})}_{\text{control weighting}} + \Delta_{\text{alloc}} + \Delta_{\text{maint}} + \Delta_{\text{agg}}. \quad (2)$$

The construction is non-circular: the per-enzyme E_i — the rising-limb Arrhenius activation energy of each enzyme’s turnover $k_{\text{cat},i}(T)$ alone (the folding term is the high-temperature deactivation, and its contribution to the rising E is negligible, -0.008 eV control-weighted) — are model inputs, and the result is how the organism value aggregates and departs from them. The growth flux-control coefficients $\hat{C}_i = \partial \ln \mu / \partial \ln k_{\text{cat},i}$ are computed by symmetric perturbation and re-solve and normalised over the controlling set. The **allocation** term is isolated by disabling the growth-law coupling, the **maintenance** term by flattening NGAM(T), and the **aggregation** term is computed directly as the Sharpe–Schoolfield E of the kinetic-backbone aggregate curve minus the control-weighted per-enzyme mean. Both organisms are analysed on their calibrated model at

their operating point, each carrying its own sector/growth-law layer, with the uncoded energy side-reactions closed. For Figure 1 the calibration posteriors (the saved Markov-chain draws) were propagated through the Sharpe–Schoolfield fitter to obtain the E_a posterior distributions. Full construction, parameter tables, data provenance and robustness analyses are in the Supplement.

5 Data and code availability

All figures and tables are generated from the analysis outputs of the two etc-GEMs and their cross-organism comparison. The complete model construction for both organisms, the parameter and data-source tables, the window-sensitivity audit, the calibration diagnostics, and the robustness analyses are provided in the Supplement.

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